NVIDIA Launches Space Computing, Rocketing AI Into Orbit

NVIDIA Accelerated Computing Platforms Boost AI Applications From Earth to Space

News Summary:

- Engineered for size-, weight- and power-constrained environments, NVIDIA Space-1 Vera Rubin Module, IGX Thor and Jetson Orin platforms deliver data-center-class performance and edge AI inferencing for orbital data centers, geospatial intelligence and autonomous space operations.
- Aetherflux, Axiom Space, Kepler Communications, Planet Labs PBC, Sophia Space and Starcloud are using NVIDIA accelerated computing platforms to power next-generation space missions.

GTC—NVIDIA today announced that its latest accelerated computing platforms are unlocking a new era of space innovation, bringing AI compute to orbital data centers (ODCs), geospatial intelligence and autonomous space operations.

By bringing data-center-class performance to size-, weight- and power (SWaP)-constrained environments, NVIDIA is enabling AI applications to operate seamlessly from ground to space, and space to space, while supporting increasingly complex mission profiles.

NVIDIA Space-1 Vera Rubin Module is the latest part of the NVIDIA accelerated platform for space. Compared with the NVIDIA H100 GPU, the Rubin GPU on the module delivers up to 25x more AI compute for space-based inferencing, enabling next-generation compute for ODCs, advanced geospatial intelligence processing and autonomous space operations.

The NVIDIA IGX Thor™ and NVIDIA Jetson Orin™ platforms deliver energy-efficient, high-performance AI inference, image sensing and accelerated data processing to enable true edge computing on orbit in a compact module.

NVIDIA data center platforms, including the NVIDIA RTX PRO™ 6000 Blackwell Server Edition GPU, deliver high-throughput, on-demand ground processing for geospatial intelligence, delivering up to 100x faster performance versus legacy CPU-based batch systems when analyzing massive imagery archives.

“Space computing, the final frontier, has arrived. As we deploy satellite constellations and explore deeper into space, intelligence must live wherever data is generated,” said Jensen Huang, founder and CEO of NVIDIA. “AI processing across space and ground systems enables real-time sensing, decision-making and autonomy, transforming orbital data centers into instruments of discovery and spacecraft into self-navigating systems. With our partners, we’re extending NVIDIA beyond our planet — boldly taking intelligence where it’s never gone before.”

Bolstering Space Missions

Industry leaders Aetherflux, Axiom Space, Kepler Communications, Planet, Sophia Space and Starcloud are using NVIDIA accelerated computing platforms to power next-generation space missions across orbital and ground environments.

Baiju Bhatt, founder and CEO of Aetherflux, said: “At Aetherflux, we’re pioneering a new paradigm for power and compute in space. NVIDIA Space-1 Vera Rubin Module delivers high-performance, energy-efficient AI at the edge in orbit, powered by solar energy. This enables autonomous operations and mission-critical services, and unlocks scalable, space-based AI infrastructure beyond Earth.”

Mina Mitry, CEO of Kepler Communications, said: “Kepler Communications is building the next-generation data network that enables real-time connectivity in space. NVIDIA Jetson Orin brings advanced AI directly to our satellites, allowing us to intelligently manage and route data across our constellation and turning our network into a smarter, more efficient platform that reduces latency and delivers secure, reliable connectivity at global scale.”

Will Marshall, cofounder and CEO of Planet, said: “Planet images the Earth every day, a data challenge that requires the world's most advanced computing. By integrating NVIDIA’s accelerated platform from space to ground, we are supercharging our ability to index the physical world. Using NVIDIA CorrDiff AI models, we are moving from raw pixels to actionable insights in near real time. Together, we are enabling a revolutionary leap in planetary intelligence, helping humanity make smarter decisions at the speed of global change.”

Rob DeMillo, CEO of Sophia Space, said, “Sophia Space’s focus is on building modular, passively cooled, hosted computing platforms that give customers dedicated infrastructure to run applications directly in space. NVIDIA Jetson Orin enables us to embed AI capability into that infrastructure, supporting real-time processing and autonomous operations within strict size, weight and power constraints. This brings cloud-like flexibility to space and makes orbital computing commercially accessible."

Philip Johnston, CEO of Starcloud, said: "Starcloud is building purpose-designed orbital data centers to deliver cloud and AI infrastructure directly in space. With NVIDIA, we can bring true hyperscale-class AI computing to orbit — processing data at the source, reducing downlink dependency and enabling customers to run training and inference workloads in space for the first time. This is a critical step toward making space a seamless extension of the global cloud."

AI-Powered Infrastructure in Orbit
The rapid growth of the commercial space industry means increased demand for real-time data processing in orbit.

NVIDIA Space-1 Vera Rubin Module delivers data-center-class AI at scale, enabling large language models and advanced foundation models to operate directly in space. Its tightly integrated CPU-GPU architecture and high-bandwidth interconnect provide the performance and memory needed to process massive data streams from space-based instruments in real time. By bringing hyperscale AI capability into orbital platforms, Space-1 Vera Rubin Module unlocks on-orbit analytics, autonomous scientific discovery and rapid insight generation.

NVIDIA IGX Thor provides industrial-grade durability and enterprise software support in a power-efficient platform designed for next-generation, mission-critical edge environments. With support for real-time AI processing, functional safety, secure boot and autonomous operation, it enables spacecraft to process sensor data locally, optimize bandwidth use and enhance responsiveness — while seamlessly complementing and extending the capabilities of ground control systems.

NVIDIA Jetson Orin delivers high-performance AI inference in an ultra-compact, energy-efficient module built for edge deployment. Optimized for SWaP-constrained environments, it enables real-time processing of vision, navigation and sensor data directly onboard spacecraft, reducing latency and optimizing bandwidth.

The NVIDIA Jetson™ platform's AI software ecosystem and CUDA® acceleration make Jetson Orin ideal for satellites, on-orbit servicing vehicles and space-based sensing platforms that require intelligent, responsive computing while remaining integrated with ground operations.

NVIDIA Data Center Platforms Advance Geospatial Intelligence
As the space ecosystem expands, so does the amount of data it generates. While on-orbit compute increases the real-time processing capabilities of geospatial sensing satellites such as imaging sensors, radars and radio frequency sensors, much of the collected data will join hundreds of petabytes of historical archive on Earth to support large-scale geospatial trend analysis.

Ground-based geospatial imaging processing systems have historically run on CPUs, resulting in longer computational turnaround times. The NVIDIA RTX PRO 6000 Blackwell Server Edition GPU delivers massive acceleration for on-ground processing over traditional architectures.

In addition, by harnessing the flexibility of CUDA, geospatial intelligence customers can adapt their processing across the cloud, edge ground stations and on orbit. They can also rapidly incorporate new AI capabilities and dynamically extract insights from these massive imagery archives for:

- **Disaster Response and Environmental Monitoring:** AI-accelerated processing of high-resolution imagery enables immediate identification of wildfires, floods and oil spills to trigger rapid alerts.
- **Climate and Weather Predictions:** Agile and precise tracking of weather patterns and long-term climate changes enables advanced analytics on atmospheric data.
- **Infrastructure and Resource Management:** Automating complex object detection and trend analysis powers the autonomous monitoring of global energy grids, transport networks and agricultural health.

Availability
The NVIDIA IGX Thor and Jetson Orin platforms, along with the NVIDIA RTX PRO 6000 Blackwell Server Edition GPU, are available today, with NVIDIA Space-1 Vera Rubin Module to be available at a later date.

Watch the GTC keynote from Huang and explore physical AI, robotics and vision AI sessions.

About NVIDIA
NVIDIA (NASDAQ: NVDA) is the world leader in AI and accelerated computing.

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